Integral Approach to the Geometry of Circles, Spheres, and Pi

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1 Introduction to the Concept of the Integral

To introduce the concept of the integral, we consider a curve. To find the area bounded by intervals a and b, we subdivide it into a number of rectangles approaching infinity

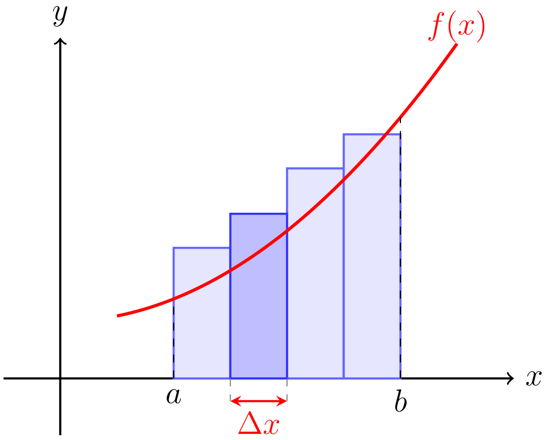

Figure 1: Approximation of the area under a curve using rectangles, where Δx represents the width of a subinterval.

$$S = \lim_{\Delta x \to 0} \left(\sum f(x_i) \Delta x \right) \tag{1}$$

Subsequently, the expression $S = \lim_{\Delta x \to 0} \left(\sum f(x_i) \Delta x \right)$ will be written as $\int_a^b f(x)\, dx$. where $f(x_i)$ represents the function evaluated at point x_i, and dx denotes an infinitesimally small change in x.and is referred to as an integral. While the previously shown expression is a definite integral, the process of finding an indefinite integral is known as antidifferentiation, which serves as the inverse operation of differentiation.

2 Application of the Integral to the Equation of a Circle

To derive the general formula for the area of a circle, we consider the equation $y = \sqrt{r^2 - x^2}$. Integrating this expression from 0 to the radius r and multiplying the result by 4 yields the total area:

$$S = 4 \int_0^r \sqrt{r^2 - x^2}\, dx \tag{2}$$

To proceed with the derivation, we apply the substitution $x = ru$, which yields:

$$dx = r\, du$$

Next, we factor out r^2 from the integrand:

$$S = 4 \int_0^1 \sqrt{r^2 - (ru)^2} \cdot r\, du = 4r^2 \int_0^1 \sqrt{1 - u^2}\, du$$

Finally, the remaining integral expression is defined as a constant named Pi (π), the computation of which will be detailed in Section 5 and 6. This yields the final expression for the area of a circle:

$$S = \pi r^2 \tag{3}$$

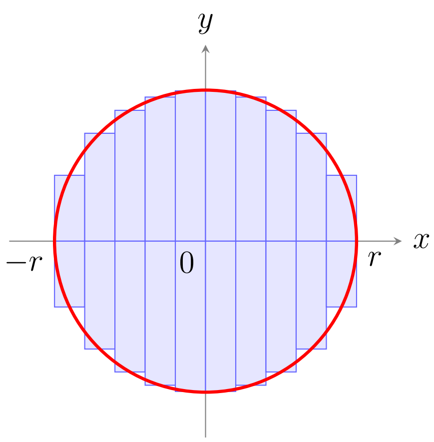

Figure 2: Partitioning a circle into vertical rectangles of equal width to approximate its total area.

3 Computation of Spherical Volume via Integration

To determine the volume of a sphere, we apply the method of circular cross-sections. The thickness of each disk approaches zero, while the cross-sectional radius r varies along the axis. The relationship between the cross-sectional radius and the sphere's radius R is governed by the equation of a circle, where $r^2 = R^2 - x^2$. Consequently, the area of an inscribed cross-section is given by $\pi(R^2 - x^2)$. Taking the limit as the disk thickness Δx approaches zero yields the integral expression for the volume of the sphere:

$$V = \lim_{\Delta x \to 0} \left(\sum \pi(R^2 - x^2)\Delta x \right)$$

$$V = \pi \int_{-R}^{R} (R^2 - x^2)\, dx \tag{4}$$

To evaluate the resulting integral, we apply the Fundamental Theorem of Calculus. Calculating the antiderivative of each term gives:

$$V = \pi \left(\int_{-R}^{R} R^2\, dx - \int_{-R}^{R} x^2\, dx \right)$$

where $F'(x) = R^2 \implies F(x) = R^2 x$, and $F'(x) = x^2 \implies F(x) = \frac{x^3}{3}$. Substituting the integration limits $-R$ and R into the expressions yields:

$$V = \pi\left[(F(R) - F(-R)) - (F(R) - F(-R))\right] = \pi \left[\left(R^3 + R^3\right) - \left(\frac{R^3}{3} + \frac{R^3}{3} \right) \right]$$

$$V = \pi \left(2R^3 - \frac{2R^3}{3} \right) = \pi \left(\frac{6R^3}{3} - \frac{2R^3}{3} \right)$$

This results in the final formula for the volume of a sphere:

$$V = \frac{4}{3}\pi R^3 \tag{5}$$

4 The Binomial Theorem and Fractional Powers

The binomial expansion represents a sequence of coefficients generated when a sum of terms is raised to the n-th power:

$$(a+b)^n = \sum_{k=0}^{n} \frac{n!}{k!(n-k)!} a^{n-k} b^k = a^n + na^{n-1}b + \frac{n(n-1)}{2!} a^{n-2} b^2 + \dots \quad (6)$$

These coefficients correspond to the entries of Pascal's triangle, where each row represents the powers of the expansion starting from $n = 0$:

$$\begin{array}{lccccccccc}
n=0: & & & & & 1 & & & & \\
n=1: & & & & 1 & & 1 & & & \\
n=2: & & & 1 & & 2 & & 1 & & \\
n=3: & & 1 & & 3 & & 3 & & 1 & \\
n=4: & 1 & & 4 & & 6 & & 4 & & 1
\end{array}$$

If the exponent n in the binomial theorem is a fraction (e.g., $1/2$), the expansion yields an infinite series rather than a finite expression. Interestingly, the coefficients of this series can still be conceptually interpolated between the rows of Pascal's triangle. Applying this rule to the function $\sqrt{1-x^2}$ yields the following expansion:

$$\sqrt{1-x^2} = 1 - \frac{1}{2}x^2 - \frac{1}{8}x^4 - \frac{1}{16}x^6 - \frac{5}{128}x^8 - \dots \quad (7)$$

Using the binomial series expansion for $\sqrt{1-u^2}$, the constant Pi (π) can be computed by integrating from 0 to $1/2$ and subtracting the area of the triangle:

$$12 \cdot \left[\int_0^{1/2} \left(\sum_{k=0}^{\infty} \binom{1/2}{k} (-1)^k u^{2k} \right) du - \frac{\sqrt{3}}{8} \right] = \pi$$

In this sequence, as the number of terms k approaches infinity, the approximation yields higher precision for the constant π. For instance, the partial sums evaluate as follows:

$$\begin{aligned}
k = 1 &\implies 2.408 \\
k = 3 &\implies 3.091 \\
k = 5 &\implies 3.123 \\
k = 15 &\implies 3.14159263
\end{aligned}$$

5 Computational Architecture and Performance Analysis of Pi Algorithms

To proceed with the precise calculation of Pi (π), high-performance computing resources are required, utilizing the C programming language. Constructing an application capable of evaluating π up to 1000 decimal places necessitates the integration of the `gmp.h`, `time.h`, and `stdio.h` libraries. Furthermore, determining the exact number of iterations required to guarantee this target precision relies heavily on the convergence rate of the proposed series:

$$\log_{10}(4) \approx 0.602 \implies \left\lceil \frac{1000}{0.602} \right\rceil = 1661 \text{ iterations} \tag{8}$$

For a comprehensive evaluation, a comparative analysis was conducted against the world's fastest algorithm for computing Pi, which operates on the principles of modular functions and elliptic curves. The empirical results demonstrated a substantial performance disparity:

- **Proposed method execution time:** 165.69 ms
- **Chudnovsky method execution time:** 0.12 ms

The computational benchmarks were executed on a MacBook Pro M3 equipped with 8 GB of unified memory. The complete source code is provided supplementary to this PDF document. To visualize these characteristics, the convergence behaviors of both frameworks are illustrated in Figure 3.

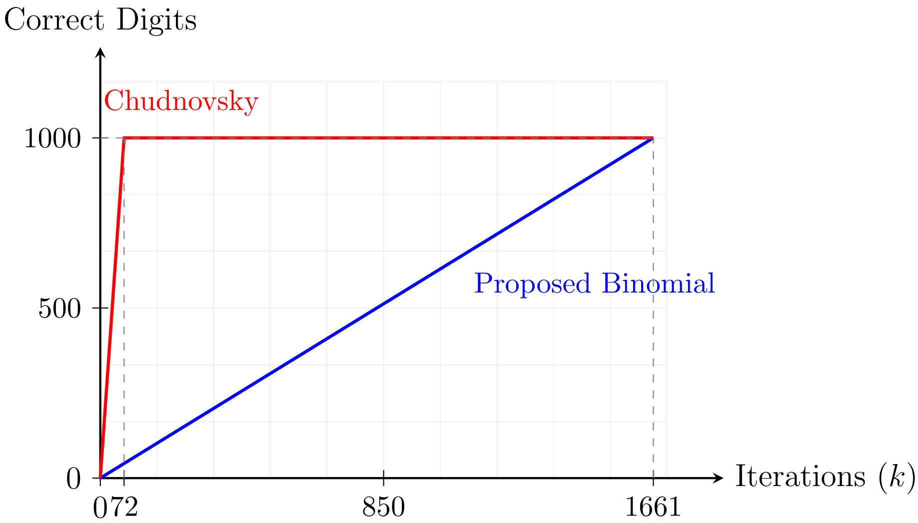

Figure 3: Algorithmic convergence chart mapping the accumulation of correct decimal digits against the total iteration count (k).

References

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